

EyeXplain Autism: Interactive System for Eye Tracking Data Analysis and Deep Neural Network Interpretation for Autism Spectrum Disorder Diagnosis

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ABSTRACT

Over the past decade, Deep Neural Networks (DNN) applied to eye tracking data have seen tremendous progress in their ability to perform Autism Spectrum Disorder (ASD) diagnosis. Despite their promising accuracy, DNNs are often seen as 'black boxes' by physicians unfamiliar with the technology. In this paper, we present EyeXplain Autism, an interactive system that enables physicians to analyse eye tracking data, perform automated diagnosis and interpret DNN predictions. Here we discuss the design, development and sample scenario to illustrate the potential of our system to aid in ASD diagnosis. Unlike existing eye tracking software, our system combines traditional eye tracking visualisation and analysis tools with a data-driven knowledge to enhance medical decision-making for physicians.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

Explainable AI, Eye Tracking, Deep Neural Network, Visualization, Autism Diagnosis

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1 INTRODUCTION

Autism Spectrum Disorder (ASD) is a complex life-long neurodevelopmental disorder often characterised by deficits in social, emotional and cognitive skills. While 'gold standard' assessment tools are available [22, 23], they are often costly, time-consuming and can be subjective. There is a need to develop new methods that

augment existing clinical tools in order to reduce waiting periods for access to care. This is critical because early intervention can provide long-term improvements and greater effect on outcomes for the child [10].

There is a recent uptake on the use of data-driven approaches to automate medical diagnosis [37]. In the context of ASD diagnosis, computer vision and Deep Neural Networks (DNN) have been shown to be useful for the quantification of behavioural/biological markers that can further lead to a non-invasive, objective and automatic tool for ASD research [6]. A considerable amount of research has been applied to eye tracking data [1, 5, 7, 8, 11–13, 16, 17, 19–21, 25, 27, 31, 34–36].

Despite the increasing accuracy of DNN models, there is a need to open the 'black box' by providing physicians a better understanding on how models generate their final decisions [18]. This will help build trust and confidence in model's predictions, potentially augmenting existing clinical approaches for a faster and more efficient ASD screening.

In this paper, we present EyeXplain Autism, a interactive system for eye tracking data analysis and interpretation of DNN predictions for autism diagnosis. Unlike existing eye tracking software, our system augments statistical graphs and typical visualisations with a data-driven knowledge to enhance medical decision-making for physicians.

2 SYSTEM DESIGN AND IMPLEMENTATION

This section is split into three parts: (i) DNN for feature extraction and Support Vector Machine (SVM) for ASD diagnosis, (ii) EyeXplain Autism Interface and (iii) a sample scenario to illustrate the potential use of our system in a clinical setting. The system uses a Python back-end and a web-based interface built using HTML, CSS, Javascript and Plotly.

2.1 Deep Neural Network for Autism Diagnosis

Our system implements a DNN for feature extraction (Figure 1 **A**) and an SVM for autism diagnosis (Figure 1 **B**). This section introduces the dataset used, eye tracking procedure, model architecture, training protocols and results.

2.1.1 Dataset. The eye tracking study included 17 ASD children and 17 typically developing (TD) children. The ASD participants were recruited as part of the Australian Autism Biobank follow-up project by the Cooperative Research Centre for Living with Autism [9]. They were matched with TD children by age (4.6 ± 0.5 years old) at the time of the study. All participants in the ASD

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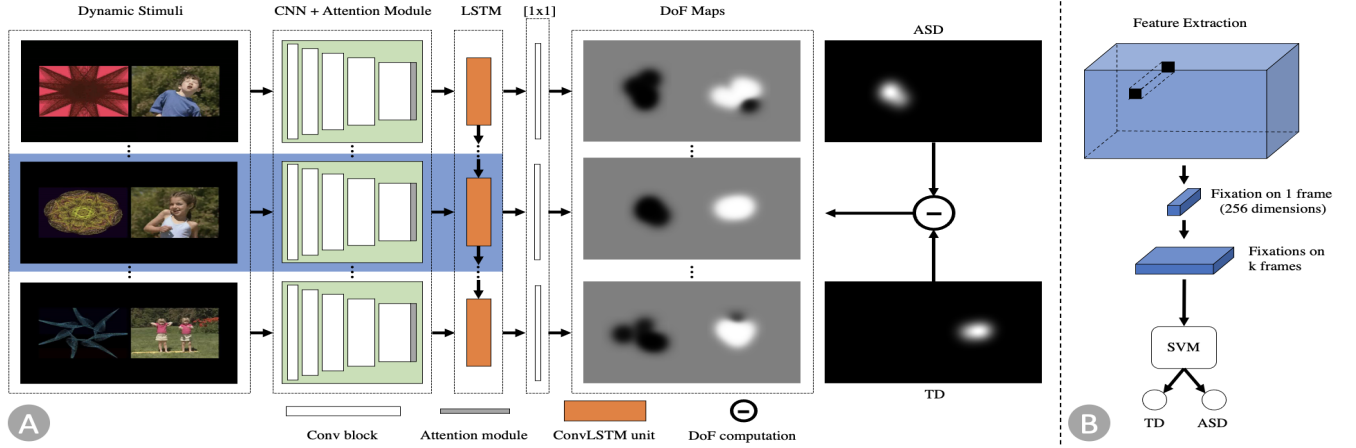


Figure 1: Overview of the proposed feature learning and autism diagnosis approach. A Given a video input, per-frame features are learned using an end-to-end approach to predict DoF maps; **B** Extracted features at fixated pixels from each fixation stage are cascaded and passed on to an SVM to identify individuals with ASD and TD.



Figure 2: Screenshots showing three frames of the dynamic stimuli (GeoPref test) used: It contains moving geometric images (left side) and moving social images (right side)

group met criteria for ASD based on the DSM V [15] and diagnosis was confirmed using the Autism Diagnostic Observation Schedule (ADOS), Second Edition [22]. There is no specific exclusion criteria for the ASD group in this study. For the TD group, exclusion criteria included known neurodevelopmental disorders, significant developmental delays and known visual/hearing impairments.

2.1.2 Eye Tracking Procedure. Participants were tested using the Tobii X2-60 eye tracker. Eye movements were recorded at 60 Hz (with an accuracy of 0.5°) during the dynamic stimuli viewing. Each participant was seated approximately 60 cm in front of a 22" monitor with a video resolution of 1680 x 1050 pixels in a quiet room. A built-in five-point calibration in Tobii Studio was completed before administering the task for accurate tracking. Tobii Studio's I-VT filter [28] was used to process the raw eye-tracking data, exclude random noise and define variables (e.g. fixations and saccades) for further analysis. More specifically, short fixations (<100 ms) were discarded and adjacent fixations (75ms, 0.5°) were merged.

2.1.3 Dynamic Stimuli. The GeoPref Test dynamic stimulus (Figure 2), which has been shown to be an effective stimulus for detecting ASD subgroups [26, 29, 30] was used. This stimulus consists of moving geometric images (MGIs) on the left side and moving social images (MSIs) on the right side. The MGIs were constructed from a collection of animated screen saver recordings. The MSIs were constructed from a series of short videos of children performing

yoga exercises. It included video recordings of children performing a wide range of movements (e.g. waving arms and dancing). The stimulus contained a total of 28 different scenes.

2.1.4 Model Architecture. As shown in Figure 1A, ACLNet [33], one of the best models available for dynamic saliency detection, is used for feature extraction. It consists of a combination of Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) network trained with an attention mechanism to enable rapid, end-to-end saliency prediction. Since ACLNet already contains an attention network trained on TD individuals, we trained our model with Difference of Fixation (DoF) maps that highlight more fixations of the TD group. This resulted in better training performance compared to a model trained on DoF maps that highlight more fixations of the ASD group. In particular, let I^+ and I^- be the fixation maps for the ASD and TD groups, respectively. The DoF map of an image is similar to [17]:

$$D = \frac{1}{1 + e^{-I/\sigma_I}} \quad (1)$$

where $I = I^- - I^+$ is a pixel-wise subtraction of ASD and TD fixation maps and σ_I represents the standard deviation of I . The resulting per-frame DoF maps encode the difference in visual attention of ASD and TD individuals. The white regions of the DoF map illustrate the visual attention of TD individuals while the black regions are for ASD individuals.

2.1.5 Training Protocol. Our model was optimised using the following loss function [14] that considers three different saliency evaluation metrics. We denote the predicted per-frame DoF map as $Y \in [0, 1]^{28 \times 28}$ and the ground truth per-frame DoF map as $Q \in [0, 1]^{28 \times 28}$. In particular, our loss function combines Kullback-Leibler (KL) divergence, the Linear Correlation Coefficient (CC) and the Normalised Scanpath Saliency (NSS):

$$L = L_{KL} + 0.1L_{CC} + 0.1L_{NSS} \quad (2)$$

L_{KL} is widely used for training saliency models computed by:

$$L_{KL}(Y, Q) = \sum_x Q(x) \log \frac{Q(x)}{Y(x)} \quad (3)$$

L_{CC} measures the linear relationship between Y and Q :

$$L_{CC}(Y, Q) = -\frac{cov(Y, Q)}{\sigma(Y)\sigma(Q)} \quad (4)$$

where $cov(Y, Q)$ is the covariance of Y and Q while σ is the standard deviation. L_{NSS} is defined as:

$$L_{NSS}(Y, Q) = -\frac{1}{N} \sum_x \bar{Y}(x) \times Q(x) \quad (5)$$

where $\bar{Y} = \frac{Y - \mu(Y)}{\sigma(Y)}$ and $N = \sum_x Q(x)$. It computes the mean of scores from the normalised per-frame DoF maps ($\bar{Y}$) at the predicted per-frame DoF maps Y .

We iteratively train our model with sequential DoF maps and video frames. We apply a loss defined over the predicted dynamic DoF maps from convLSTM. Let $\{Y_t^d\}_{t=1}^T$ and $\{Q_t^d\}_{t=1}^T$ denote the predicted dynamic DoF maps and continuous DoF maps. We minimise the following loss:

$$L^d = \sum_{t=1}^T L(Y_t^d, Q_t^d) \quad (6)$$

The parameters of ACLNet are initialized to the pre-trained parameters [33], then fine-tuned on the current dataset.

2.1.6 ASD Classification. Once the model has been trained to predict per-frame DoF maps of ASD and TD individuals from the given dynamic stimulus, feature extraction and classification are performed (Figure 1B). Based on the eye tracking data, we determine the fixation positions and the corresponding frames in which they are recorded. Each saccade-fixation pair was considered as a fixation stage. For each fixation stage, features are extracted from the corresponding fixation position on the feature map obtained from the convLSTM output (note: convLSTM output is upsampled 4 times before extracting the feature map). More specifically, given a frame where a fixation has been identified, the feature map at the corresponding fixation is extracted, which results in a 256-dimensional feature vector. For a corresponding number of fixation stages, feature vectors for all fixations are concatenated in their temporal order starting from the first fixation to the last fixation stage. This serves as the feature space in which classification is performed. If there were fewer number of identified fixations, zeros are appended to the end. A linear decision boundary between ASD and TD individuals was determined by training an SVM on the extracted features.

2.1.7 Implementation Details. We implemented our model in Tensorflow with Keras and Scikit-learn libraries. For the training phase, we fine-tuned the network with Adam optimizer and a batch size of one image for a total of 20 epochs. The learning rate was set to 0.0001. There is no dropout and data augmentation performed. L2 regularisation with the penalty parameter $C = 1$ was used for SVM classification.

2.1.8 Evaluation Metrics and Results. We report the performance of our model in terms of accuracy, sensitivity (i.e. true positive rate) and specificity (i.e. true negative rate). In the context of ASD diagnosis, these metrics have been widely used. To eliminate bias

in the estimate of the probability of error [32], we used a leave-one-subject-out cross-validation. For our experiments, we fixed the weights of the fine-tuned DNN and perform SVM on top of the extracted features. In addition, we explored the optimal number of fixation stages, starting from 1 fixation to 30 fixations. Our model's performance on different fixation stages are as follows: accuracy: 68%-100%, sensitivity: 57%-100% and specificity: 65%-100%. For deployment, we set the number of fixation stages to 13, the value in which the highest accuracy has been reported. The SVM coefficients learned during the last fold are stored in a server for later processing.

2.1.9 Computational Load. The entire training procedure for feature learning takes about 1 hour with two NVIDIA RTX 2080 Super and a 3.5GHz Intel CPU (i7-7800X). Once the DNN model has been trained, feature extraction and SVM classification can be performed in less than 1 minute. The trained DNN model and SVM coefficients are then stored in a server and later accessed by the EyeXplain Autism Interface using a Python webframework, CherryPy.

2.2 EyeXplain Autism Interface

The EyeXplain Autism Interface consists of four components as shown in Figure 3: A control panel at the top A, a visualisation tab B with an interactive timeline at the bottom C and a statistics tab E.

2.2.1 Control Panel. The control panel A allows users to upload a new eye tracking data and dynamic stimulus used during the experiment. It also allows users to switch between the visualisation and statistics tabs for task-specific analysis. By clicking the 'Sort Fixations' button, users can sort fixations by feature importance from the SVM output. As discussed in the previous section, each recorded fixation is represented by 256 values. Since we set the value of fixation stages to be 13, there are 13×256 values as input to the SVM and the same number of coefficients are learned by the SVM. To sort the fixations by feature importance, we compute for the sum of the 256 coefficients from each frame, which results in 13 values. When these values are sorted in descending order, the frames with highest influence to diagnosis can be extracted. Finally, users can perform automated diagnosis shown in a popup box by clicking the 'Predict' button.

2.2.2 Visualisation Tab. The visualisation tab B enables users to play the dynamic stimulus, to scrub the timeline to jump to a different frame, to view recorded fixations B2 and to turn on/off heatmap visualisation of ASD and/or TD samples that were seen by the DNN during training. The heatmaps are encoded using different colours, where ASD heatmap B1 is red while TD heatmap B3 is blue.

2.2.3 Interactive Timeline. The interactive timeline C shows the chronological sequence of fixations recorded in the uploaded eye tracking data. Each fixation is represented as a square thumbnail (100x100 pixels), centred at the fixation point (x,y) recorded in a given frame. When the mouse is hovered over a thumbnail, the interface shows a comparison popup C1, which displays randomly chosen ASD and TD fixations that the DNN has seen on that same

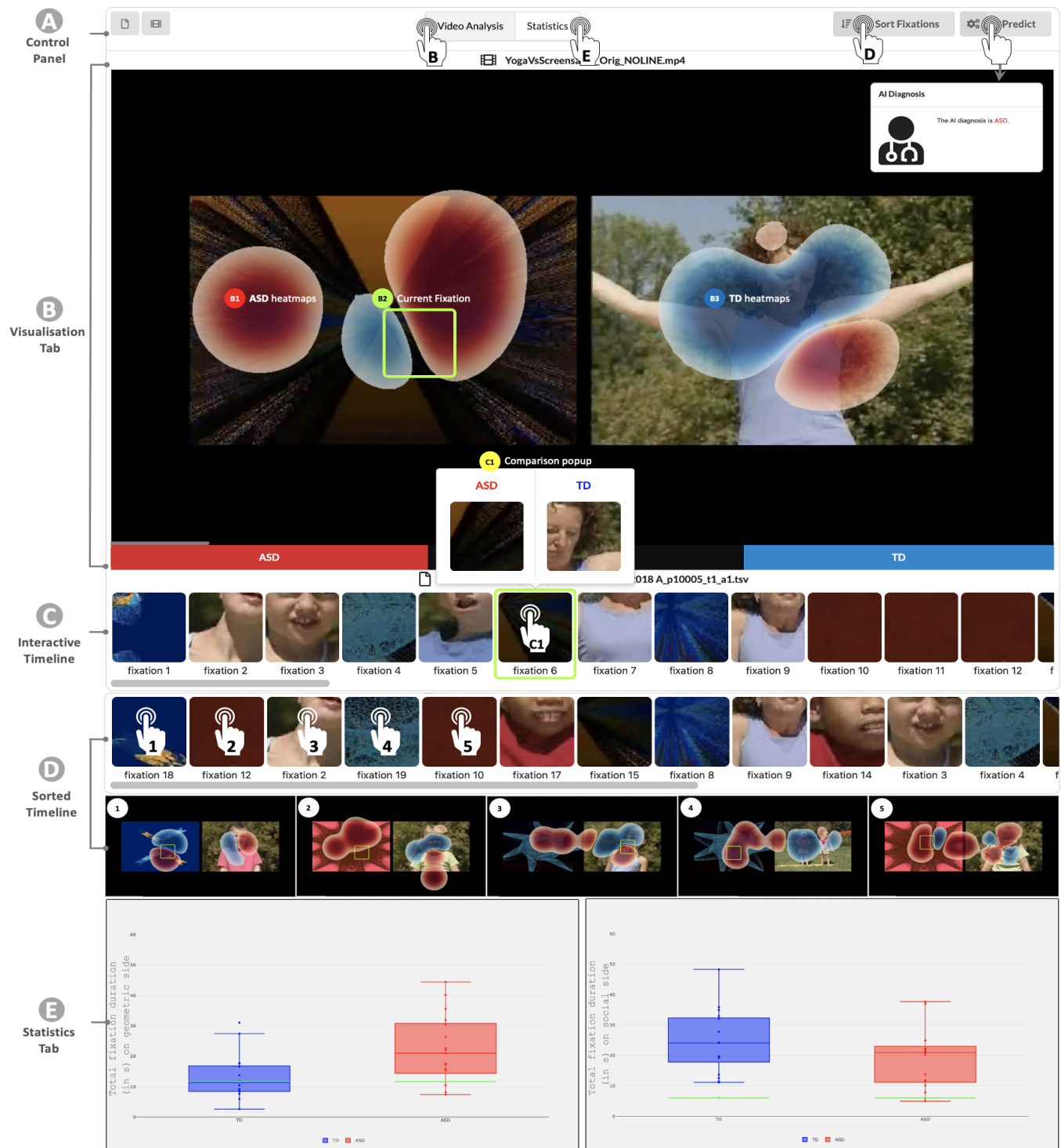


Figure 3: EyeXplain Autism Interface. The control panel **A** allows users to upload eye tracking data and dynamic stimulus, switch between visualisation and statistics tabs, sort fixations and perform diagnosis. The visualisation tab **B** shows a frame with recorded fixation **B2** and ASD **B1** and TD **B3** heatmaps. Each recorded fixation is represented as a thumbnail in the interactive timeline **C**. Hovering over any fixation displays a comparison popup **C1**. The 'Sort Fixations' button sorts feature importance **C** → **D**. The 'Predict' button shows a popup with the predicted diagnosis. The statistics tab shows different variables for quantitative analysis **E**.

frame during training. When a thumbnail is clicked, the visualisation tab will show the frame where the corresponding fixation was recorded.

2.2.4 Statistics Tab. The statistics tab **E** allows users to quantitatively analyse eye tracking data using box plots. For our dynamic stimulus, we defined two areas of interest (AOI): (i) geometric (left) side of the stimulus and (ii) social (right) side of the stimulus. Using our dataset, we pre-computed variables, such as fixation count, fixation duration and percentage of duration, around these two AOIs and saved them for later comparison. When a new eye tracking data is loaded, these variables are shown to help users compare them with variables computed from the new eye tracking data.

2.2.5 User Interaction Performance Test. In order to ensure interactivity and reduce latency, all computationally intensive processing (e.g. feature extraction, SVM classification, feature importance computation) are handled by the backend server. In exploratory user tests, the DNN model latency is at ~ 0.1 sec/prediction. On the other hand, the computation of SVM feature importance is performed within ~ 0.1 seconds.

2.3 Sample Scenario

We present a sample scenario showing how our system can help physicians better analyse eye tracking data and understand DNN predictions for ASD diagnosis (illustrated in the supplementary file). Our physician Hubble is studying a new eye tracking data collected from his patient. He decides to use the EyeXplain Autism interface to screen his patient for autism. Using the control panel **A**, he uploads a new eye tracking data and selects the dynamic stimulus used during the experiment. He sees the visualisation tab **B** where he plays the dynamic stimulus and observes fixation locations recorded during the experiment.

2.3.1 How different is the current fixation to the aggregate ASD and TD samples? Hubble starts by selecting a fixation of interest (fixation6 in the interactive timeline **C**) and activating the ASD and TD heatmap visualisation in the visualisation tab **B**. He finds that the majority of ASD samples **B1** fixated on the geometric side of the screen, while the majority of the TD samples **B3** fixated on the woman shown on the screen.

2.3.2 How different is the current fixation to an ASD and a TD sample? Hubble is interested to determine the similarity between the current fixation and individual ASD and TD fixations seen during training. In this case, he hovers to a fixation of interest (fixation6 in the interactive timeline **C**). The comparison popup **C1** shows that the current fixation is similar to a fixation previously recorded from an ASD sample. Hubble observes that the current fixation is different to a representative TD sample where a fixation was recorded on the face of the woman.

2.3.3 What is the DNN's diagnosis? Why is the prediction like that? Hubble clicks the 'Predict' button in the control panel **A** and sees a popup appear showing that the DNN has determined that the current sample has ASD. Hubble now wants to know why the DNN prediction was ASD. He clicks the 'Sort Fixations' button in

the control panel **A** to determine which frames are most important for diagnosis. The interactive timeline has now been updated and sorted by feature importance from the SVM output **C** $\rightarrow$ **D**. Upon interacting with the sorted timeline **D** and inspecting the top five fixations (labeled 1-5 in **D**) and their corresponding visualisation tab, Hubble notices that the top five recorded fixations fall within/near the ASD heatmaps, which helps explain why the DNN output was ASD.

2.3.4 How does the DNN output compare with quantitative analysis? After knowing and interpreting the DNN prediction, Hubble wants to know if quantitative analysis will reveal similar prediction. He switches to the statistics tab **E** to analyse the eye tracking data quantitatively. In particular, he is interested in two variables: (i) total fixation duration on the geometric side ($duration_{geo}$) and (ii) total fixation duration on the social side ($duration_{soc}$). The statistics tab displays these two variables with corresponding distribution of ASD (in red) and TD (in blue) samples seen during training. In addition, the current sample is highlighted as a green line. Upon analysis, he observes that the $duration_{geo}$ of the current sample is within the distribution of both ASD and TD samples. On the other hand, the $duration_{soc}$ of the current sample is lower than TD samples and within the distribution of ASD samples. He sequentially analyses the other variables (e.g. ratio of ($duration_{geo}$) to total duration) and finds out that the current sample has eye tracking variable values that are within the distribution of ASD samples seen during training.

2.3.5 Outcome. By using our interactive system to interpret DNN's output and quantitatively analyse eye tracking data, Hubble is convinced that the current sample has ASD. Standard clinical tests have revealed that the current sample indeed has ASD. Satisfied with the result, Hubble is now interested to use our system in his clinic for ASD screening.

3 LIMITATIONS AND ONGOING WORK

3.1 Small Dataset

Despite promising results, our DNN is currently trained on a relatively small dataset of 17 ASD and 17 TD children. We plan to collect more data to ensure reliability and robustness of autism diagnosis. Since multiple factors (e.g. age, content and dynamic nature of the stimuli) affect visual attention in children [24], we plan to build a collection of dynamic stimuli and determine the most discriminative videos that highlight differences in visual attention of ASD and TD participants. This would allow for a more efficient screening system.

3.2 DNN Architecture

Over the past decade, DNNs that predict visual attention have seen tremendous leap in performance [4]. While there exists a number of dynamic saliency models, we chose ACLNet[33] and did not explore other models for our system. The main rationale is that ACLNet has consistently held first place in all performance measures in several benchmark datasets. Furthermore, ACLNet has an attention module that enables rapid, end-to-end saliency prediction. We plan

to build a new saliency model, compare it with existing models and determine the suitable model for feature extraction.

3.3 Autism Severity and Treatment Response Prediction

We are working on extending the capability of our system to predict autism severity and treatment response. This would help determine which patients would have positive response to intervention, thereby allowing for more effective, efficient and even targeted treatments.

3.4 EyeXplain Autism Interface Design and Evaluation

We consulted the eye-tracking literature (see [2, 3] for review) and followed design recommendations for DNN-assisted medical systems [37] to establish the initial design and supported features of our interface. We will conduct co-design workshops with physicians to iterate the interface design, add more features and evaluate its effectiveness. Our goal is to determine if our system improves model's interpretability, user trust, confidence, and usability in a clinical setting.

4 CONCLUSION

In this paper, we presented EyeXplain Autism, an interactive system we are developing that visualises eye tracking data, enables an interpretable DNN for ASD diagnosis, and supports quantitative analysis commonly used in eye tracking studies. By presenting a sample scenario, we showed the potential of our system to help physicians analyse new eye tracking data, compare it to previously seen eye tracking data, perform automated ASD diagnosis, better interpret predictions of complex DNN models and build trust towards their predictions.

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